

AWS IAM CLOUD SECURITY PROJECT

Project overview

I concluded a project on Amazon Web Services (AWS). This project aimed to build and manage a basic cloud infrastructure using Amazon Web Services (AWS). The goal was to gain hands-on experience with identity management, compute resources, storage services, and activity monitoring within the AWS environment.



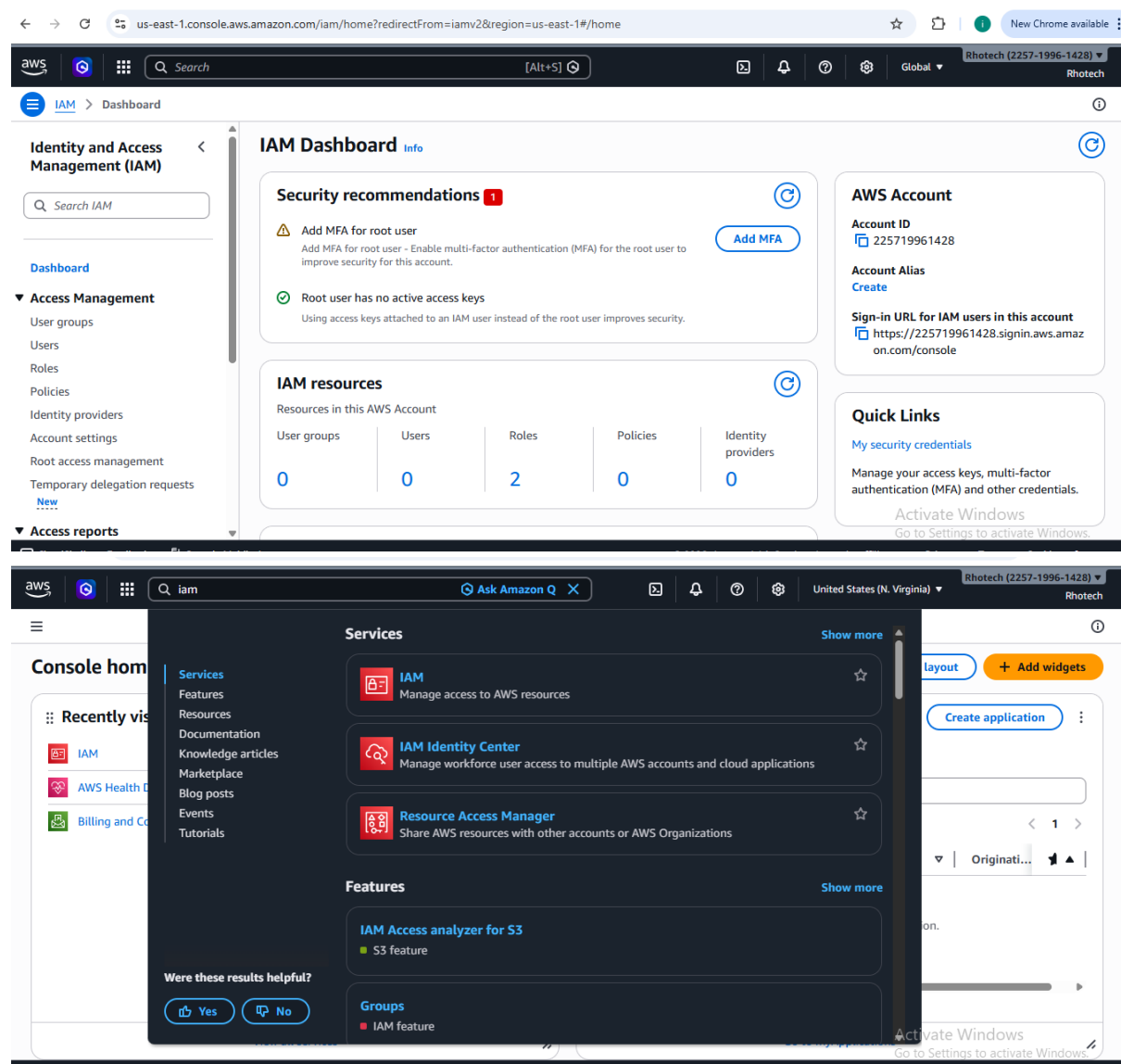
I created an account on aws.amazon.com, first setting up the root account and configuring Identity and Access Management (IAM), defining roles to grant users access to the AWS Management Console. Rhotech-IAM-Roland was created and given administrator access

Task 1.

Identity and Access Management Configuration

The first phase of the project involved configuring secure user access through **AWS Identity and Access Management (IAM)**. An IAM group was created to represent a specific role within the environment, allowing permissions to be managed collectively. A user account was then created and added to the group. A custom policy was attached to the group to define the permissions granted to the user, ensuring controlled and role-based access to AWS resources. Below is a picture demonstration.

Diagram



aws [Search] [Alt+S] Ask Amazon Q Global Rhotech (2257-1996-1428) Rhotech

Identity and Access Management (IAM)

User groups (0) Info

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

Search

Group name Users Permissions Creation time

No resources to display

Create group

Access Management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management

Temporary delegation requests

New

Access reports

CloudShell Feedback Console Mobile App

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Specify user details

Step 1: Specify user details

Step 2: Set permissions

Step 3: Review and create

Step 4: Retrieve password

User details

User name

Rhotech-IAM-Roland

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and +, -, @, _ (hyphen)

Provide user access to the AWS Management Console - optional

In addition to console access, users with `SignInLocalDevelopmentAccess` permissions can use the same console credentials for programmatic access without the need for access keys.

Console password

Autogenerated password

You can view the password after you create the user.

Custom password

Enter a custom password for the user:

Must be at least 8 characters long

Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + - (hyphen) = [] { }

Show password

Users must create a new password at next sign-in - Recommended

Users automatically get the `IAMUserChangePassword` policy to allow them to change their own password.

If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user.

Learn more

Activate Windows

Go to Settings to activate Windows.

aws [Search] [Alt+S] Ask Amazon Q Global Rhotech (2257-1996-1428) Rhotech

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

Add user to group

Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

Copy permissions

Copy all group memberships, attached managed policies, and inline policies from an existing user.

Attach policies directly

Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1449)

Choose one or more policies to attach to your new user.

Search

Filter by Type

All types

Policy name Type Attached entities

☒ AccessAnalyzerServiceRolePolicy AWS managed 0

☐ AccountManagementFromVeracode AWS managed 0

☒ AdministratorAccess AWS managed - job function 0

☐ AdministratorAccess-Amplify AWS managed 0

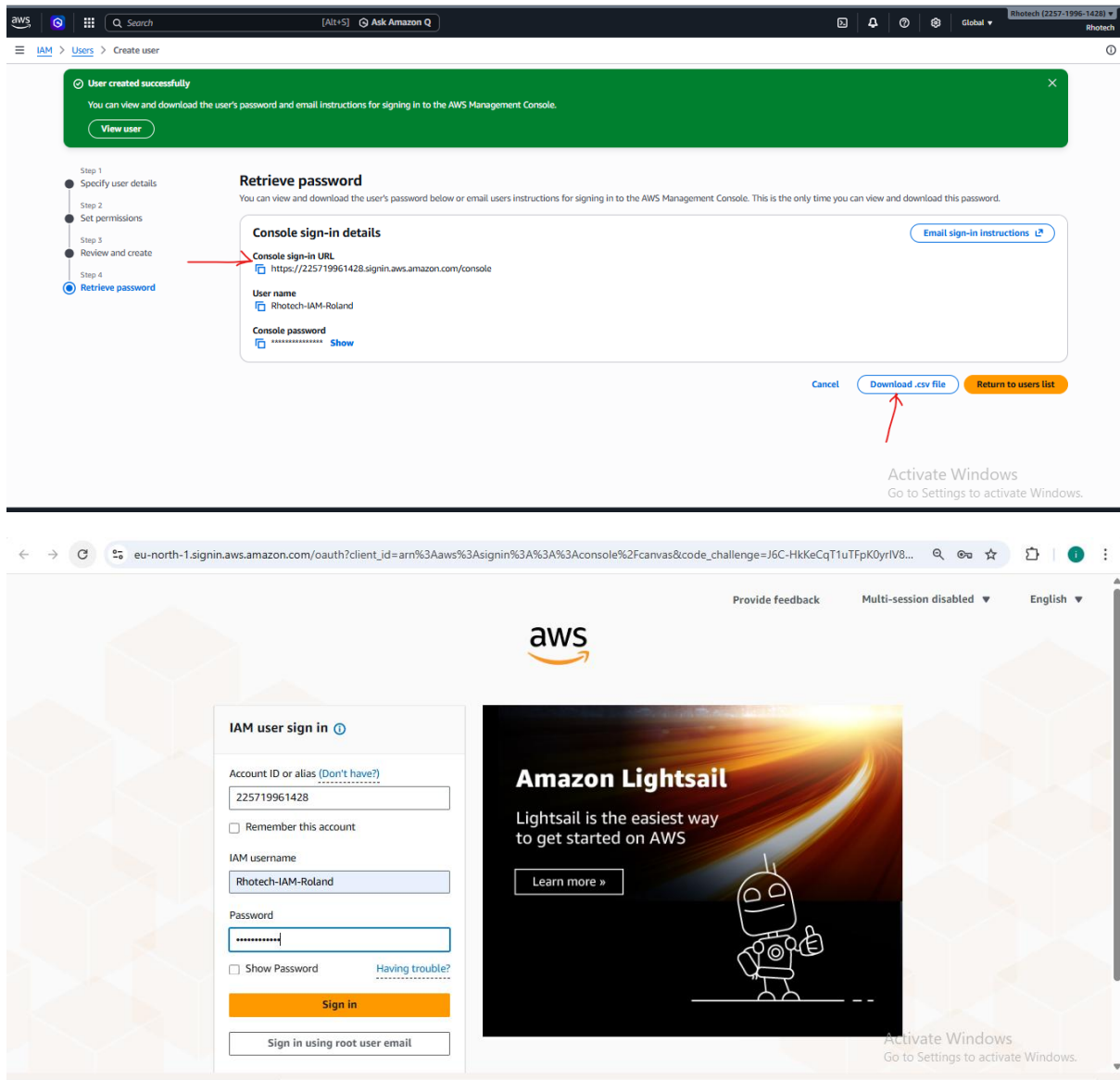
☐ AdministratorAccess-AWSElasticBeanstalk AWS managed 0

☐ AIOPsAssistantIncidentReportPolicy AWS managed 0

Create policy

Activate Windows

Go to Settings to activate Windows.



In the above diagram, I created a user and set the permission as administrator.

Task 2

After creating the IAM user and assigning the policy, I created two groups and a member for my project (Rhotech-IAM-IT, Rhotech-IAM-Sale). Each member was assigned to a group. Members were added to the group by clicking on the group and checking the box to apply the change.

The image displays two screenshots of the AWS IAM console, illustrating the process of creating a user.

Top Screenshot: IAM Users List

The top screenshot shows the AWS IAM console's "Users" page. The left sidebar contains the "Identity and Access Management (IAM)" menu. The main content area shows a list of users. A red arrow points to the "Create user" button in the top right corner.

User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key ID	Active key age
Rhotech-IAM-IT-Jack	/	1	-	-	1 minute	-	-	-
Rhotech-IAM-Roland	/	0	19 hours ago	-	19 hours	1 hour ago	-	-

Bottom Screenshot: Create User Wizard

The bottom screenshot shows the "Create user" wizard, specifically the "Specify user details" and "Set permissions" steps.

Specify user details:

- User name:** Rhotech-IAM-IT-Betty
- Provide user access to the AWS Management Console - optional:** Checked
- Console password:** Custom password (entered: *****)
- Users must create a new password at next sign-in - Recommended:** Checked

Set permissions:

- Permissions options:** Add user to group (selected)
- User groups (1/2):**

Group name	Users	Attached policies	Created
Rhotech-IAM-IT	1	-	2026-03-04 (22 hours ago)
Rhotech-IAM-Sale	0	-	2026-03-04 (20 hours ago)

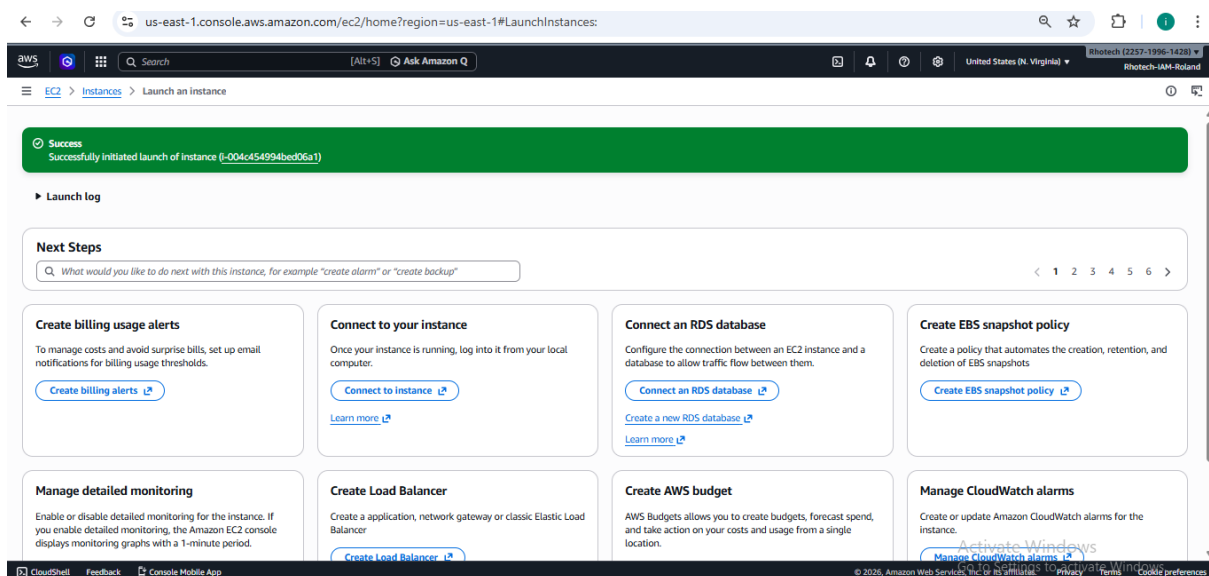
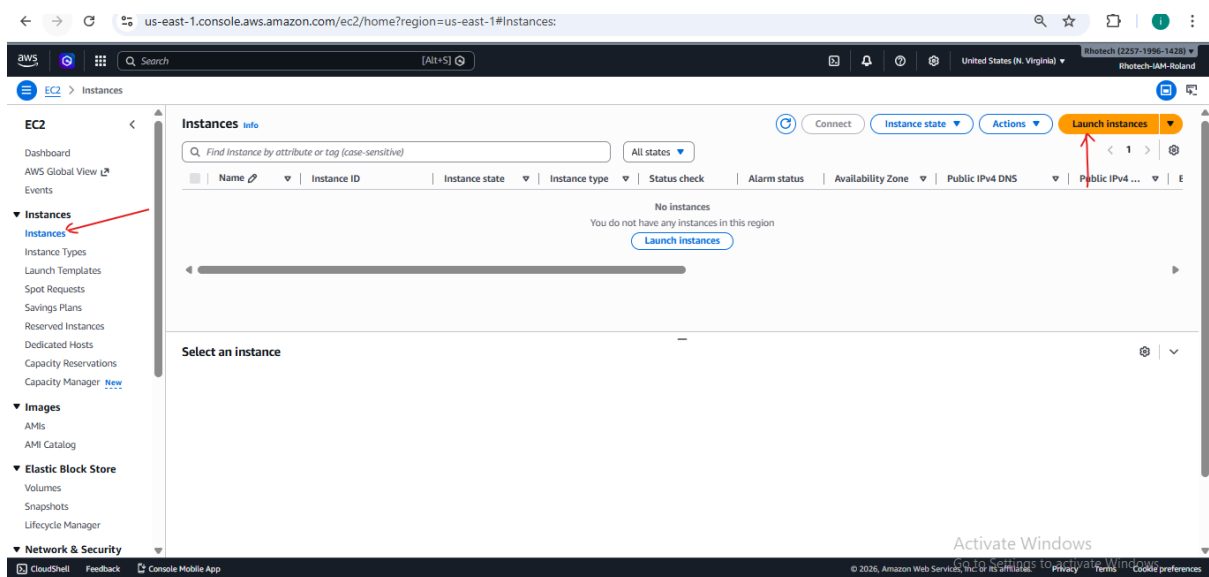
Red arrows in the bottom screenshot point to the "Add user to group" option, the "Rhotech-IAM-IT" group in the list, and the "Next" button.

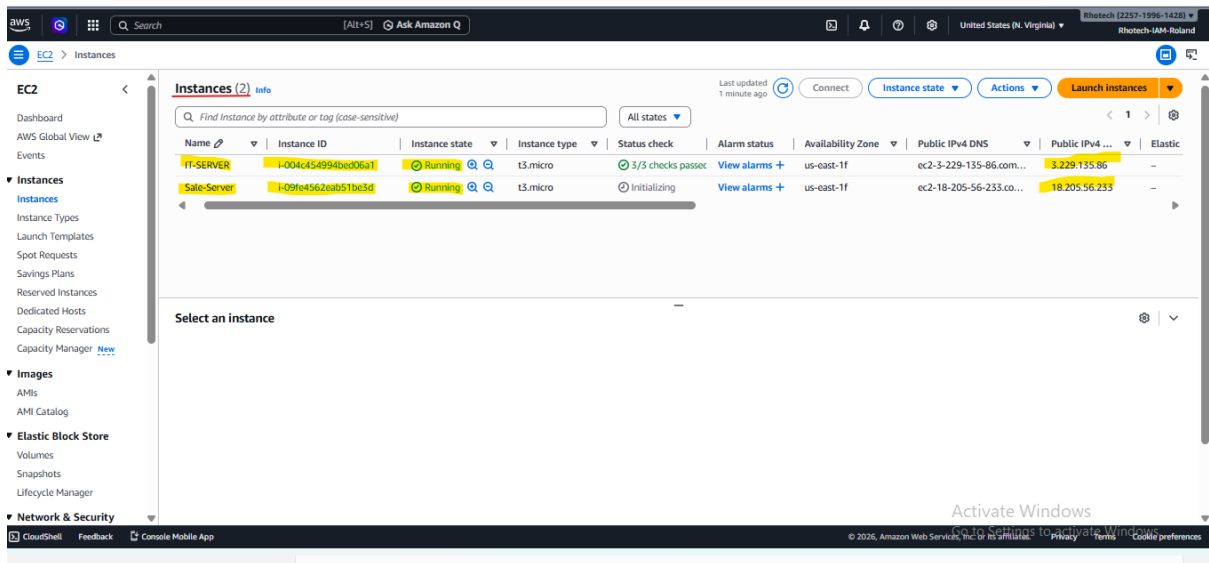
The picture above shows how groups and users are created.

Task 3.

Another service I used after creating the User, Group, and Member is the Elastic Compute Cloud (EC2), also known as instances. EC2 provides virtual servers in the cloud.

In the image below, I set up my instances by clicking on Launch Instances. I also assigned a name and selected appropriate options for Amazon Machine Image (AMI), Key Pair, Instance type, and network settings.

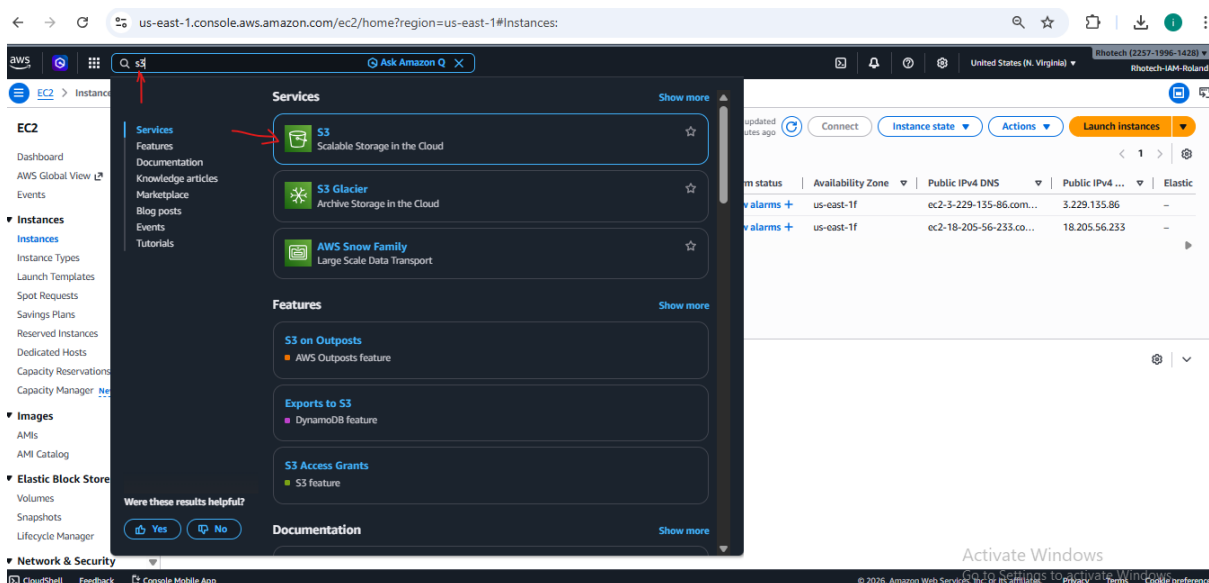


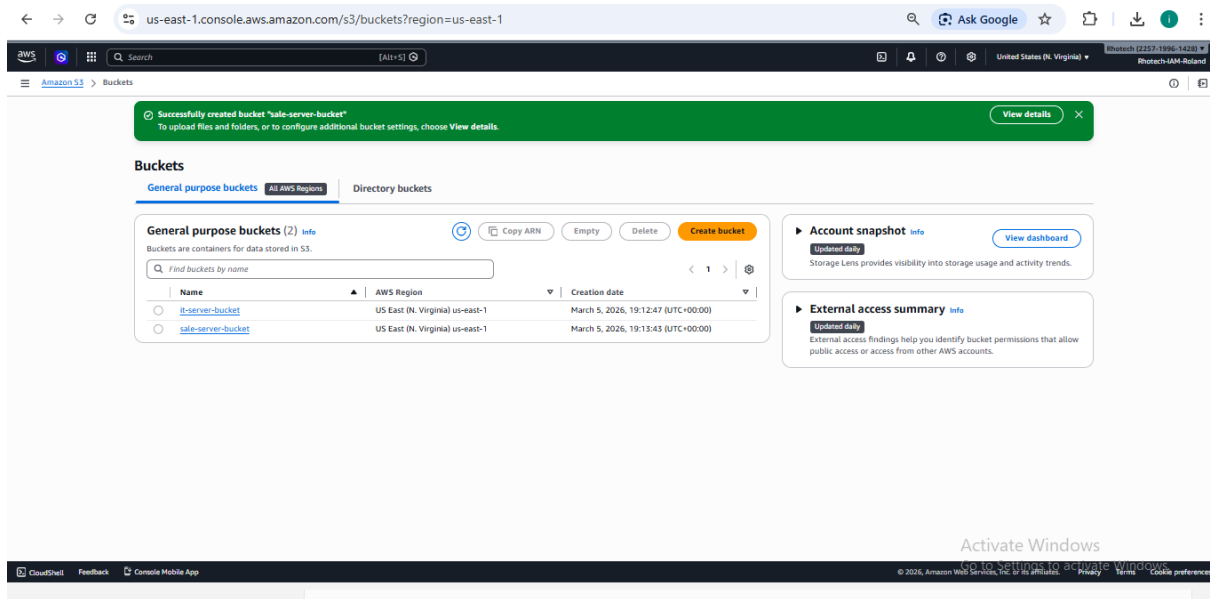


The picture above shows the step-by-step guide on how I set up the EC2

Task 4.

In this task, I set up scalable cloud storage using an S3 bucket, creating the IT-Server Bucket and the Sales-Server Bucket. In this task, I also block public access to my database. The encryption type was selected: (Server Side Encryption with Amazon S3 managed keys (SSE-S3)). In this project, I selected the encryption type based on the project's requirements.

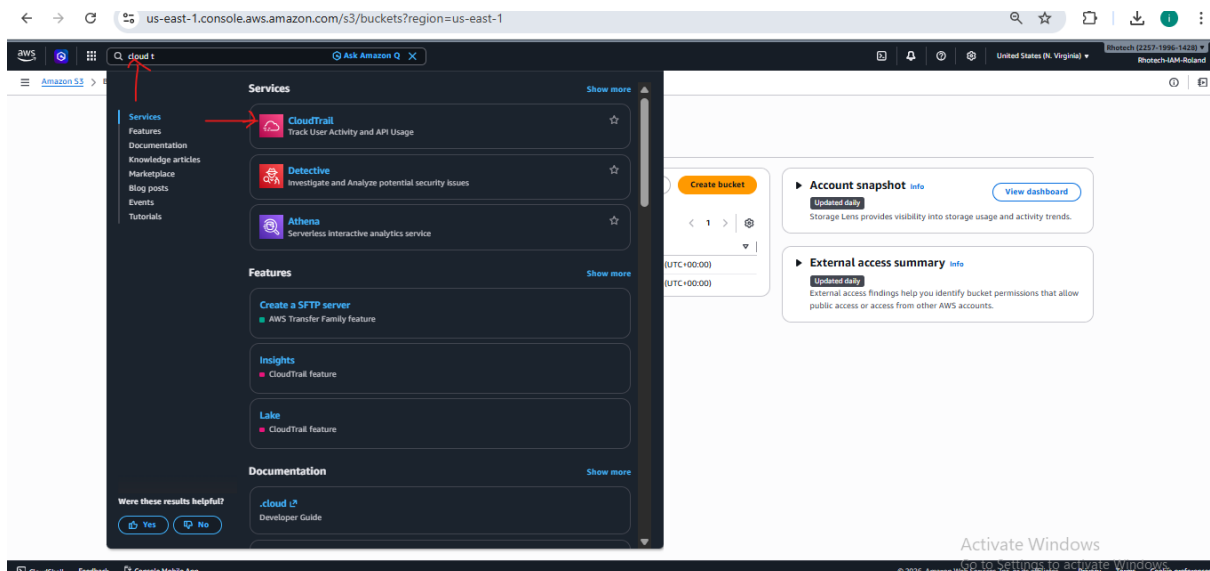


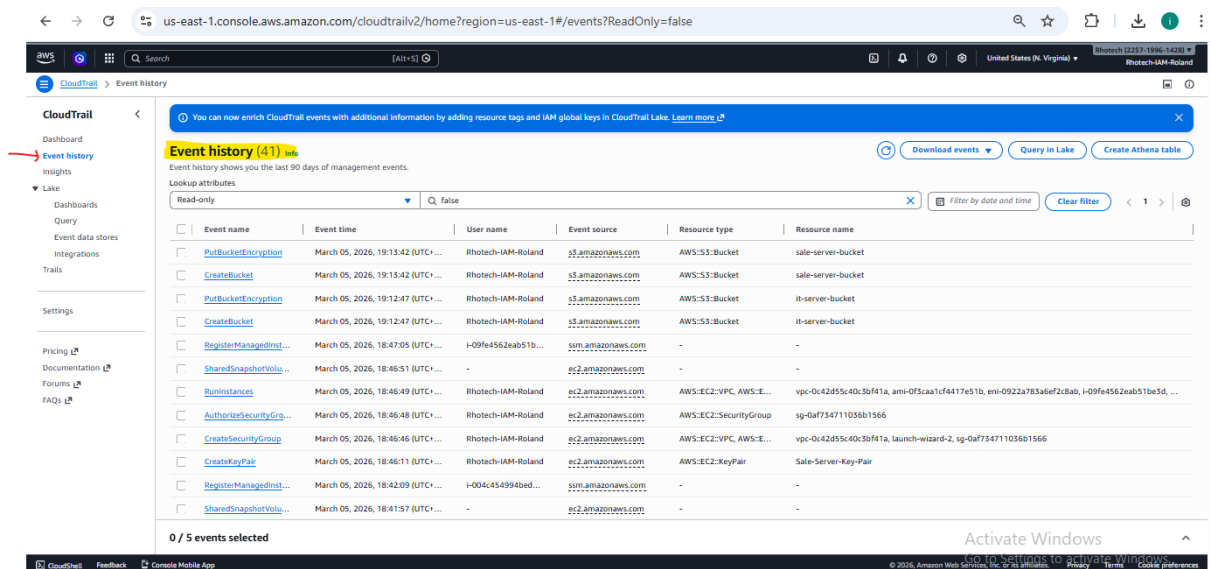
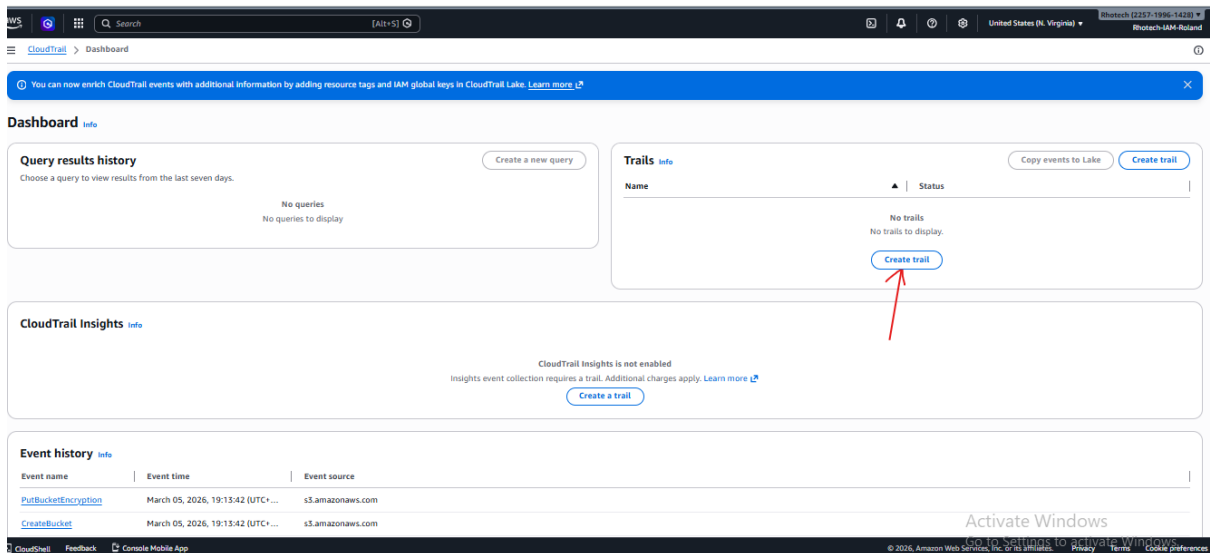


Task 5.

In this task, I set up a CloudTrail.

The purpose of this task is to track user activity and API usage. Also, the event history shows 90 days of management, data, insight, and network activity events.





The image above shows how event history works in CloudTrail.

Task 6.

User Permission

In this task, I assign a policy to the IAM user; this policy is specific to each user in the group. The purpose of this permission is to define privileged access. This also aligns with the principle of least privilege, as shown in the pictures below.

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/details/Rhotech-IAM-IT-Betty?section=permissions

Identity and Access Management (IAM)

Dashboard

Access Management

Roles

Policies

IAM users

IAM user groups

Identity providers

Account settings

Root access management

Temporary delegation requests

Access reports

Access Analyzer

Resource analysis

Unused access

Analyzer settings

Credential report

Organization activity

Service control policies

Resource control policies

amaws:iam::225719961428:user/Rhotech-IAM-IT-Betty

Enabled without MFA

Created March 05, 2026, 15:21 (UTC)

Last console sign-in Never

Create access key

Permissions

Groups (1)

Tags

Security credentials

Last Accessed

Permissions policies (0)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type

Search

Policy name

Type

Attached via

No resources to display

Remove

Add permissions

Add permissions

Create inline policy

Permissions boundary (not set)

Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this user, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

Generate policy

No requests to generate a policy in the past 7 days.

Permissions options

☐ Add user to group

Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions

Copy all group memberships, attached managed policies, inline policies, and any existing permissions boundaries from an existing user.

☒ Attach policies directly

Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1489)

Filter by Type

Search s3

25 matches

Policy name	Type	Attached entities
☐ AmazonDMSRedshiftSSRole	AWS managed	0
☐ AmazonS3ExpressFullAccess	AWS managed	0
☐ AmazonS3ExpressReadOnlyAccess	AWS managed	0
☐ AmazonS3FilesClientFullAccess	AWS managed	0
☐ AmazonS3FilesClientReadOnlyAccess	AWS managed	0
☐ AmazonS3FilesClientReadWriteAccess	AWS managed	0
☐ AmazonS3FilesCSIDriverPolicy	AWS managed	0
☐ AmazonS3FilesFullAccess	AWS managed	0
☒ AmazonS3FilesReadOnlyAccess	AWS managed	0

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/details/Rhotech-IAM-IT-Betty/add-permissions

Step 1

Add permissions

Step 2

Review

Review

The following policies will be attached to this user. [Learn more](#)

User details

User name

Rhotech-IAM-IT-Betty

Permissions summary (1)

Name	Type	Used as
AmazonS3FilesReadOnlyAccess	AWS managed	Permissions policy

Cancel

Previous

Add permissions